RAIRE And AWAIRE IRV Ranked-Choice Audit Boundaries

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Abstract

Instant-runoff voting outcomes are not audited by replaying one plurality winner-loser margin. RAIRE constructs a set of NEN and NEB assertions whose joint truth implies the reported IRV winner; AWAIRE studies adaptive weighting of such assertions during an IRV audit. This paper documents the current RCOUNT V.19 surface. The implementation records a RAIRE or AWAIRE method id, a reported IRV elimination order, ranked sample choices, and ranked-choice assertion records, then verifies that this declared evidence is well formed and replayable as a boundary contract. It does not yet generate RAIRE assertions, run AWAIRE adaptive logic, replay assertion-level risk, or certify an IRV tabulation. The result is a small but explicit record-shape contract for later ranked-choice audit math.

1 Introduction

Instant-runoff voting (IRV) changes the audit problem. A plurality audit can usually name a reported winner, a reported loser, and one outcome assertion. In an IRV contest, later-round transfers and eliminations mean that a simple winner-loser plurality assertion does not by itself imply the reported winner. The audit must instead explain why the reported elimination path and final winner are forced by the ranked ballots under the declared rule.

RAIRE addresses that problem by generating a set of assertions for the reported IRV outcome. The usual assertion families are "not eliminated before" (NEB) and "not eliminated next" (NEN). Their joint truth is chosen so that, if all assertions pass under the audit evidence, the reported IRV winner follows. AWAIRE targets the same ranked-choice setting but adaptively weights assertions as audit evidence accumulates.

RCOUNT V.19 is not yet a full RAIRE or AWAIRE implementation. It lands the record surface needed before that math can be replayed: method identity, the reported IRV elimination order, ranked sample choices, and ranked-choice assertion records. The current verifier checks that this evidence is shaped well enough to preserve and replay as a boundary transcript.

2 Assertion Contract

A full RAIRE/AWAIRE audit record must make the implication from assertion set to IRV winner public. RCOUNT currently records the first layer of that contract: which ranked-choice method is being claimed, which candidates appear in the reported elimination order, which assertions the package declares, and which ranked choices support each sampled assertion value.

Field	Meaning
`method_id`	Versioned ranked-choice audit method. Current ids are `raire-irv-v1` and `awaire-irv-v1`.
`rcv_elimination_order`	Reported IRV elimination order, with the reported winner last in the synthetic boundary fixture.
`assertions`	Declared ranked-choice assertions. The current kind is `ranked-choice-assertion`.
`assorter_id`	Assertion family or replay id, such as an NEB-style RAIRE id or an AWAIRE adaptive id.
`sample_steps`	Sampled ranked ballots or CVRs tied to an assertion id and an assorter value.
`ranked_choices`	Ordered candidate choices observed for the sampled unit.

The intended mathematical contract is stronger than the implemented one. A RAIRE record should identify NEN and NEB assertions and show that their joint truth implies the reported IRV winner. An AWAIRE record should additionally make the adaptive weighting and assertion-risk path replayable. V.19 records those claims as explicit boundary evidence, but it does not yet prove the IRV-winner implication or recompute assertion statistics.

3 Implemented Surface

The core verifier names the ranked-choice design check `verify_ranked_choice_audit_design`. It is called for every `AuditAlgorithmRun`. For non-ranked methods, the check rejects stray `rcv_elimination_order` values or nonempty `ranked_choices`; ranked evidence must not leak into unrelated audit methods. For `RAIRE_IRV_METHOD_ID` and `AWAIRE_IRV_METHOD_ID`, whose string values are `raire-irv-v1` and `awaire-irv-v1`, the check requires a ranked surface rather than a plurality surface.

The ranked method must have at least two candidates in the elimination order, at least one assertion, and every assertion must have kind `RankedChoiceAssertion`. The elimination order cannot contain duplicate candidates. Each sample step must carry nonempty ranked choices, and each choice must name a candidate from the elimination order without duplication within the sampled ranking. The generic audit-run verifier then continues with ordinary assertion id, sample-step id, rational-value, and source-reference checks.

Check	Failure caught
`is_supported_audit_algorithm_method`	Unknown method ids before ranked-choice design is considered.
`verify_ranked_choice_audit_design`	Missing or duplicate elimination-order candidates, missing ranked assertions, wrong assertion kind, missing ranked choices, unknown ranked choices, and duplicate choices within one ranking.
`audit_algorithm_transcript`	Accepted boundary records for RAIRE and AWAIRE method identity and evidence shape.
`replay_audit_algorithm_statistics`	Boundary transcripts stating that RAIRE assertion replay and AWAIRE adaptive replay are recorded but not statistically replayed.

In `rcount-audit`, `replay_audit_algorithm_statistics` dispatches both ranked methods to boundary transcripts. The RAIRE message says that IRV assertion replay is recorded but not replayed. The AWAIRE message says that IRV adaptive replay is recorded but not replayed.

4 Fixtures

The positive RAIRE fixture is `synthetic_raire_boundary_package`. It is built by `synthetic_ranked_choi` with method id `raire-irv-v1`, run id `audit-run:raire-irv-boundary`, and assorter id `raire-neb-not-eliminated-before-v1`. The synthetic reported elimination order is `cand-c`, `cand-b`, `cand-a`. It has one ranked-choice assertion, two ranked sample steps, and decision `boundary`. The corresponding unit test verifies that the package passes and preserves the elimination order.

The positive AWAIRE fixture is `synthetic_awaire_boundary_package`. It reuses the same helper and ranked evidence but names method id `awaire-irv-v1`, run id `audit-run:awaire-irv-bou` and assorter id `awaire-adaptive-irv-v1`. Its test checks that the package verifies and that the method id is the AWAIRE id.

The negative ranked-choice fixture is `synthetic_bad_raire_boundary_package`. It starts from the RAIRE boundary package and appends `cand-a` a second time to the first sampled ranking. The ranked-choice verifier rejects the package as `InvalidRankedChoiceSample`, proving that duplicate ranked choices are not accepted as replayable boundary evidence.

These fixtures are intentionally small. They test method identity, elimination order preservation, ranked sample shape, assertion kind, and malformed ranked ballot rejection. They do not test IRV tabulation, RAIRE assertion generation, assertion-level risk replay, or a real ranked CVR adapter.

5 Boundaries

The plan states the claim boundary directly: RAIRE/AWAIRE audit an RCV outcome under a reported rule and ballot record. RCOUNT now preserves ranked-choice audit evidence and reports both methods as explicit boundaries, but it does not yet generate or replay RAIRE/AWAIRE assertions.

That boundary matters because design shape and audit statistics are different claims. The implemented code can say that a RAIRE or AWAIRE record has a known method id, a nonduplicated reported elimination order, ranked-choice assertions, and sampled rankings over the declared candidates. It cannot yet say that the assertion set is sufficient, that every NEN or NEB assertion was generated correctly, that AWAIRE weights were updated correctly, or that any assertion met a risk limit.

The IRV reduction assumptions are also outside this slice. The package does not yet contain a verified ranked-choice contest model, a ranked CVR fixture, or an IRV tabulation replay proving the elimination order. Until those layers exist, V.19 should be read as a record-shape and method-identity contract, not as an RCV tabulator or a ranked-choice audit certification claim.

6 Conclusion

V.19 gives RAIRE and AWAIRE a named place in RCOUNT without overstating the implementation. The core verifier accepts `raire-irv-v1` and `awaire-irv-v1` only when the ranked-choice evidence surface is well formed: a reported elimination order, ranked-choice assertions, and sampled rankings over the declared candidates. The audit crate then reports method specific boundary transcripts rather than pretending to replay RAIRE or AWAIRE statistics.

The next work is clear from the boundary. RCOUNT needs ranked contest and CVR semantics, an IRV tabulation fixture, RAIRE assertion generation, and assertion-level risk replay before it can claim more than preservation of the ranked-choice audit record. This paper therefore records the contract that exists now and the exact line it does not cross.

References

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